Running Spark Jobs on K8s Cluster (Digital Ocean)

1. Create K8s cluster with Digital Ocean
   - add config file to local machine.
   - save it as `~/.kube/<cluster-name>-kubeconfig.yaml`
2. Install doctl and add Digital Ocean token

   ```
   snap install doctl
   doctl auth init
   ```

3. Add default cluster with doctl

   ```
   doctl kubernetes cluster kubeconfig save <cluster-name>
   ```

4. Check the get nodes to see the nodes of clusters

   ```
   kubectl get nodes
   ```

5. Now, create a project `~/Projects/K8sSpark` and add these files for test-run spark job on clusters.
   - Dockerfile

     ```
     FROM apache/spark:4.1.1

     USER root

     COPY app.py /opt/spark/app.py

     USER 185
     ```

   - app.py

     ```
     from pyspark.sql import SparkSession
     from pyspark.sql.functions import col

     spark = SparkSession.builder.appName("K8sSparkDemo").getOrCreate()

     data = [("Alice", 34), ("Bob", 45), ("Cathy", 29)]
     columns = ["name", "age"]

     df = spark.createDataFrame(data, columns)

     df_filtered = df.filter(col("age") > 30)

     df_filtered.show()

     spark.stop()
     ```

   - requirements.txt

     ```
     pyspark
     ```

6. Login Docker, Build and Push the docker-image.

   ```
   docker login

   docker build -t <docker-user-name>/<image-name>:<image-tag>

   docker push <docker-user-name>/<image-name>:<image-tag>
   ```

7. Now, get any of the k8s cluster url using the command.

   `kubectl config view --minify -o jsonpath='{.clusters[0].cluster.server}'`

8. Create a kubernetes service account and add permissions.

   ```
   kubectl create serviceaccount <k8s-service-account-name>

   kubectl create clusterrolebinding spark-role --clusterrole=edit --serviceaccount=default:<k8s-service-account-name>
   ```

9. Submit the demo spark job.

```
spark-submit \
--master k8s://https://<YOUR-API-SERVER> \
--deploy-mode cluster \
--name spark-k8s-demo \
--conf spark.executor.instances=3 \
--conf spark.executor.memory=512m \
--conf spark.driver.memory=512m \
--conf spark.kubernetes.container.image=<your-user>/<image-name>:<image-tag> \
--conf spark.kubernetes.namespace=default \
--conf spark.kubernetes.authenticate.driver.serviceAccountName=<k8s-service-account-name>\
local:///opt/spark/app.py
```

10. Check the cluster logs

```
kubectl get pods
kubectl logs <pod-name>
```